Sightings Exploration of CRAP Pinciples and concept Learnt In Class

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##INTRODUCTION

New York is one of the most popular and vibrant cities in the world, the opportunities and the beauty the city present makes it an attraction for people across the globe. However, the more people you have in a city the higher the possible of pollution, and pollution attracts rodents like RATS. The rodents thrive in urban environments, taking advantage of disposal patterns, old infrastructure, and underground transport system. This problem caused the authorities in New York city maintained a detailed record of rats sightings reported by residents.

The aim of my analysis is to:

1. Show the brough with the most Sightings over time
2. Find out the Top 7 Location Types with the most Sightings
3. Visualize the the sightings concentration using maps (sf)

Before analysis can be conducted on this data set, it is essential to clean and structure the data to ensure accuracy and consistency in the cleaning process, geospacial preparation, and visualization techniques applied to the data set while exploring the concepts learnt in class.

Install packages needed and Load dataset

```
# Install necessary packages if not installed
options(repos = c(CRAN = "https://cran.rstudio.com/"))
install.packages(c("tidyverse", "lubridate"))

## Installing packages into 'C:/Users/Password/AppData/Local/R/win-
library/4.4'
## (as 'lib' is unspecified)

## package 'tidyverse' successfully unpacked and MD5 sums checked
## package 'lubridate' successfully unpacked and MD5 sums checked

## Warning: cannot remove prior installation of package 'lubridate'

## Warning in file.copy(savedcopy, lib, recursive = TRUE): problem copying
## C:\Users\Password\AppData\Local\R\win-
library\4.4\00LOCK\lubridate\libs\x64\lubridate.dll
```

```
## to
## C:\Users\Password\AppData\Local\R\win-
library\4.4\lubridate\libs\x64\lubridate.dll:
## Permission denied

## Warning: restored 'lubridate'

##
## The downloaded binary packages are in
##  C:\Users\Password\AppData\Local\Temp\RtmpWEsoUm\downloaded_packages
```

```r
install.packages(c("tidyverse", "lubridate"))
```

```
## Installing packages into 'C:/Users/Password/AppData/Local/R/win-
library/4.4'
## (as 'lib' is unspecified)

## package 'tidyverse' successfully unpacked and MD5 sums checked
## package 'lubridate' successfully unpacked and MD5 sums checked

## Warning: cannot remove prior installation of package 'lubridate'

## Warning in file.copy(savedcopy, lib, recursive = TRUE): problem copying
## C:\Users\Password\AppData\Local\R\win-
library\4.4\00LOCK\lubridate\libs\x64\lubridate.dll
## to
## C:\Users\Password\AppData\Local\R\win-
library\4.4\lubridate\libs\x64\lubridate.dll:
## Permission denied

## Warning: restored 'lubridate'

##
## The downloaded binary packages are in
##  C:\Users\Password\AppData\Local\Temp\RtmpWEsoUm\downloaded_packages
```

```r
# Load required libraries
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3

## Warning: package 'ggplot2' was built under R version 4.4.3

## Warning: package 'dplyr' was built under R version 4.4.3

## Warning: package 'lubridate' was built under R version 4.4.3

## ── Attaching core tidyverse packages ──────────────────────── tidyverse
2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.1.5
## ✓ forcats   1.0.0     ✓ stringr   1.5.1
## ✓ ggplot2   3.5.1     ✓ tibble    3.2.1
```

```
## ✓ lubridate 1.9.4     ✓ tidyr     1.3.1
## ✓ purrr     1.0.2

## ── Conflicts ────────────────────────────────────────────
tidyverse_conflicts() ──
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```r
library(lubridate)
library(sf)
```

```
## Warning: package 'sf' was built under R version 4.4.3

## Linking to GEOS 3.13.0, GDAL 3.10.1, PROJ 9.5.1; sf_use_s2() is TRUE
```

```r
library(ggplot2)
library(dplyr)



# Load the dataset
rats_data <- read_csv("A1_sightings.csv", na = c("", "NA", "N/A"))
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame
for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)

## Rows: 101914 Columns: 52
## ── Column specification
────────────────────────────────────────────────────────
## Delimiter: ","
## chr (33): Created Date, Closed Date, Agency, Agency Name, Complaint Type,
De...
## dbl  (6): Unique Key, Incident Zip, X Coordinate (State Plane), Y
Coordinate...
## lgl (13): Landmark, Facility Type, School or Citywide Complaint, Vehicle
Typ...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

Data Cleaning needed (To Handle “NA Values” & Date formatting)

```r
# Convert 'Created Date' to proper date format
rats_data <- rats_data %>%
  mutate(created_date = mdy_hms(`Created Date`)) %>%  # Convert date column
  drop_na(created_date)  # Remove rows with missing dates
```

```r
# Extract useful date components
rats_data <- rats_data %>%
  mutate(
    year = year(created_date),
    month = month(created_date),
    day = day(created_date),
    weekday = wday(created_date, label = TRUE, abbr = FALSE)  # Full weekday
name
  )

# Clean borough names
rats_data <- rats_data %>%
  mutate(borough = str_to_title(Borough)) %>%  # Convert to Title Case
  filter(!is.na(borough))  # Remove missing boroughs

# Clean Location Type
rats_data <- rats_data %>%
  mutate(location_type = str_to_title(`Location Type`)) %>%
  replace_na(list(location_type = "Unknown"))  # Replace missing location
types

# Remove rows with missing Latitude/Longitude
rats_data <- rats_data %>%
  drop_na(Latitude, Longitude)

# Convert Latitude and Longitude to numeric
rats_data <- rats_data %>%
  mutate(
    latitude = as.numeric(Latitude),
    longitude = as.numeric(Longitude)
  )

# Remove duplicate records
rats_data <- rats_data %>%
  distinct()

# Save cleaned data to a new CSV
write_csv(rats_data, "Cleaned_A1_sightings.csv")  # Saves in the main project
folder
```

###CONCEPTS ON DATA CLEANING AND LOADING

Considering the fact that i wanted to answer those question (Business Problem) i already had an idea of the necessary libraries i would need to create my visulizations hence the reason for the libraries

Converting the date column into a standardised format was one of the most crucial tasks in getting the data ready for analysis. At first, the time of each rat sighting was entered into

the “Created Date” box, which was initially saved as a string. This format was changed to a structured date-time format since it was unsuitable for time-based analysis.

Additional time-based data, such as year, month, day, and weekday, were retrieved to aid in trend analysis. This made it possible to investigate rat activity patterns and seasonal fluctuations in more detail throughout the year.

Manhattan, Brooklyn, Queens, the Bronx, and Staten Island are the five boroughs that make up New York City. The “Borough” column had different capitalisations and formatting problems as a result of dataset inconsistencies.
Rats were recorded in residential buildings, commercial places, and public spaces, as indicated by the “Location Type” column. However, the text formatting has to be standardised due to missing data and inconsistent capitalisation. “Unknown” was used in place of missing values in order to preserve records without totally erasing valuable data.

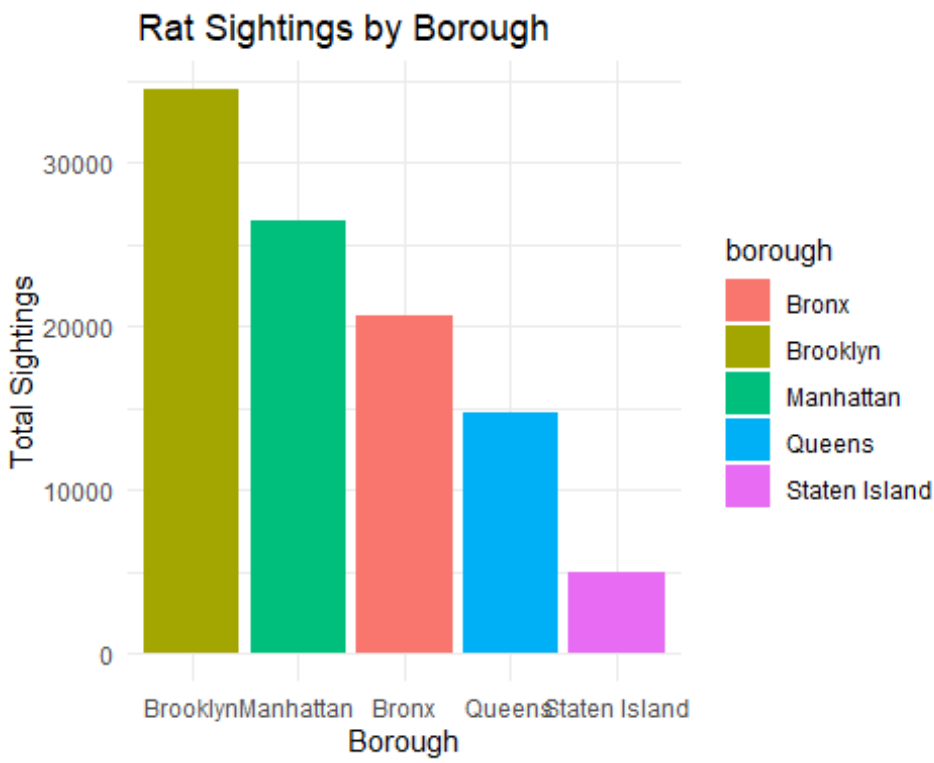
Managing Coordinates in Space It was essential to make sure that every sighting included precise geographic coordinates because this dataset will subsequently be used to map rat infestations. In order to prevent inaccurate data points on the map, missing latitude and longitude values were eliminated. In order to facilitate accurate geographic computations, latitude and longitude values were also transformed into numeric format. I aslo proceded to removing dupilcate to enhance integrity

After the geographic preparation and data cleaning were finished, the dataset was converted into a well-structured, superior resource for additional study. The following significant advancements were made:

Time-series analysis of rat sighting patterns was made possible by the standardisation of dates. To ensure consistency among records, location categories and borough names were cleansed. Rat hotspots in New York City could be mapped thanks to improvements in geospatial accuracy.

##Rats Sightings by Borough

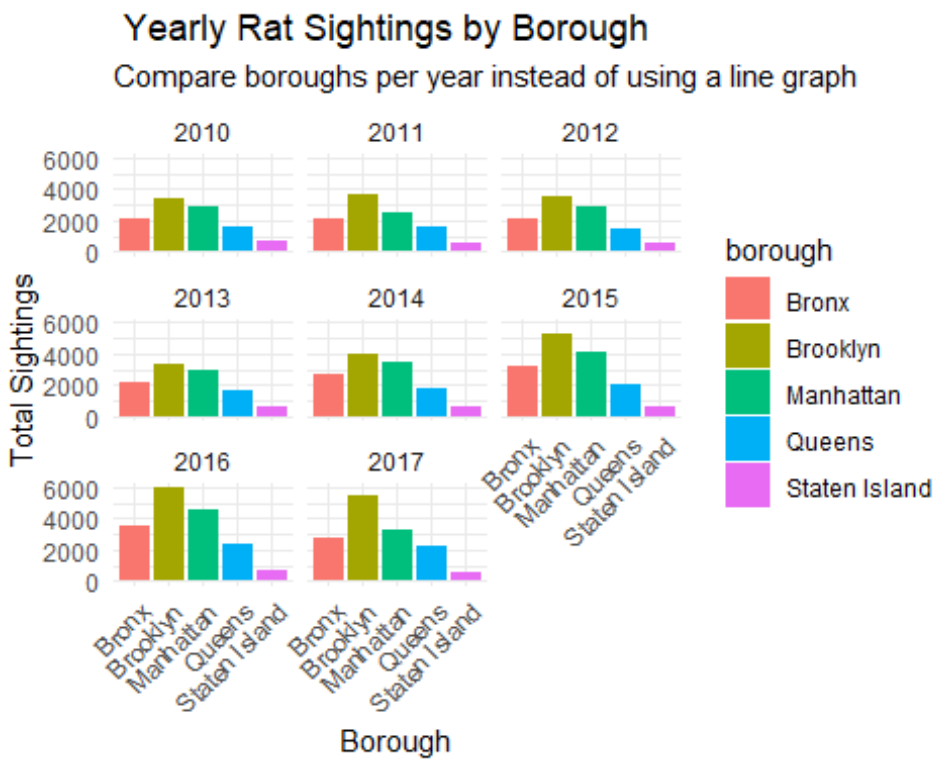
```
rats_data %>%
  group_by(borough) %>%
  summarise(total_sightings = n()) %>%
  ggplot(aes(x = reorder(borough, -total_sightings), y = total_sightings,
fill = borough)) +
  geom_bar(stat = "identity") +
  labs(title = " Rat Sightings by Borough",
       x = "Borough", y = "Total Sightings") +
  theme_minimal()
```




```r
# Save as PNG
ggsave(filename = "output/A1(1).png", width = 10, height = 7, dpi = 300)

# Save as PDF
ggsave(filename = "output/A1(1).pdf", width = 10, height = 7)
```

##Indepth Hence the reason for Facting

```r
borough_trends <- rats_data %>%
  group_by(year, borough) %>%
  summarise(total_sightings = n(), .groups = "drop")


ggplot(borough_trends, aes(x = borough, y = total_sightings, fill = borough))
+
  geom_bar(stat = "identity") +
  facet_wrap(~year) +  # Separate chart for each year
  labs(
    title = " Yearly Rat Sightings by Borough",
    subtitle = "Compare boroughs per year instead of using a line graph",
    x = "Borough",
    y = "Total Sightings"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))  # Tilt labels for
readability
```




```r
# Save as PNG
ggsave(filename = "output/A1(2).png", width = 10, height = 7, dpi = 300)

# Save as PDF
ggsave(filename = "output/A1 (2).pdf", width = 10, height = 7)
```

CONCEPT

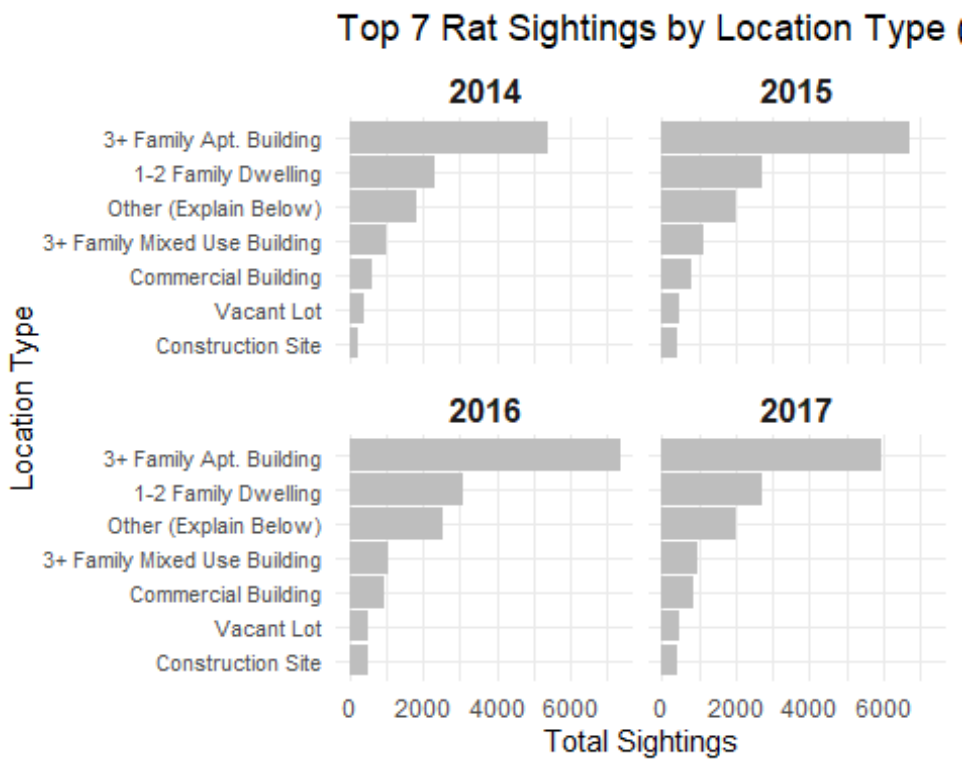
The Background of the Graphics Rat sightings by borough are compared overall in the first visualisation, which shows which places have the worst rodent infestations. By displaying how rat sightings vary in each borough over a number of years, the second visualisation expands on this approach and enables temporal comparisons. By showing which boroughs are most impacted and how infestations have changed over time, these charts collectively offer a thorough insight of NYC's rodent situation. Each borough is visually distinct through color coding, making comparisons intuitive. Repetition: Consistent use of bar charts reinforces relationships between boroughs across different time periods. Alignment: Bars are neatly aligned for easy comparison, and axis labels are rotated to improve readability. Proximity: Related data points are grouped logically, ensuring boroughs and years are easily interpretable. while also recognizing emerging trends in rodent activity. Together, these charts offer data-driven insights that can inform pest control strategies and urban sanitation efforts.

Utilising the Visualisation Principles of Healy and Cairo Boroughs can be compared not only to one another but also to different years thanks to the usage of facet wrapping.

(Comparability) Clarity & Simplicity (Cairo): By using a minimalist theme, extraneous features are eliminated, enabling the data to tell the story. Truthfulness (Cairo): By avoiding deceptive scale or representational distortions, the visualisations maintain the integrity of the data. Functionality (Cairo & Healy): Policymakers can identify new patterns in rodent activity and prioritise boroughs with ongoing infestations thanks to the clear and useful visualisations.

TOP 7 RAT SIGHTINGS BY LOCATION TYPE

```
# Filter for years 2014-2017
rats_filtered <- rats_data %>%
  filter(year >= 2014 & year <= 2017)

# Summarize sightings per location type and year, then get the top 7 per year
top7_locations <- rats_filtered %>%
  group_by(year, location_type) %>%
  summarise(total_sightings = n(), .groups = "drop") %>%
  arrange(year, desc(total_sightings)) %>%
  group_by(year) %>%
  slice_head(n = 7) #Select top 7 per year

# Create the faceted bar chart with correct formatting
ggplot(top7_locations, aes(x = total_sightings, y = reorder(location_type,
total_sightings))) +
  geom_bar(stat = "identity", fill = "grey") +  # Use black bars for
consistency
  facet_wrap(~year) +  # Create separate plots per year
  labs(
    title = "Top 7 Rat Sightings by Location Type (2014-2017)",
    x = "Total Sightings",
    y = "Location Type"
  ) +
  theme_minimal() +
  theme(
    strip.text = element_text(face = "bold", size = 12),  # Bold facet labels
    axis.text.y = element_text(size = 8)  # Improve readability of location
types
  )
```




```r
# Save as PNG
ggsave(filename = "output/A1(3).png", width = 10, height = 7, dpi = 300)

# Save as PDF
ggsave(filename = "output/A1(3).pdf", width = 10, height = 7)
```

##CONCEPTS The top 7 rat sighting sites from 2014 to 2017 are displayed in this visualisation, which shows whether infestations stay in the same areas or change over time. The chart helps in identifying recurring trouble spots, including residential structures, dining establishments, or construction sites, by facetting annually. These insights help urban planners and health officials identify high-risk areas.I made some slight changes to your original visulization by reducing the variable to 7 and using the volor grey beacuse i felt the color black was too loud to apply in that context

CRAP

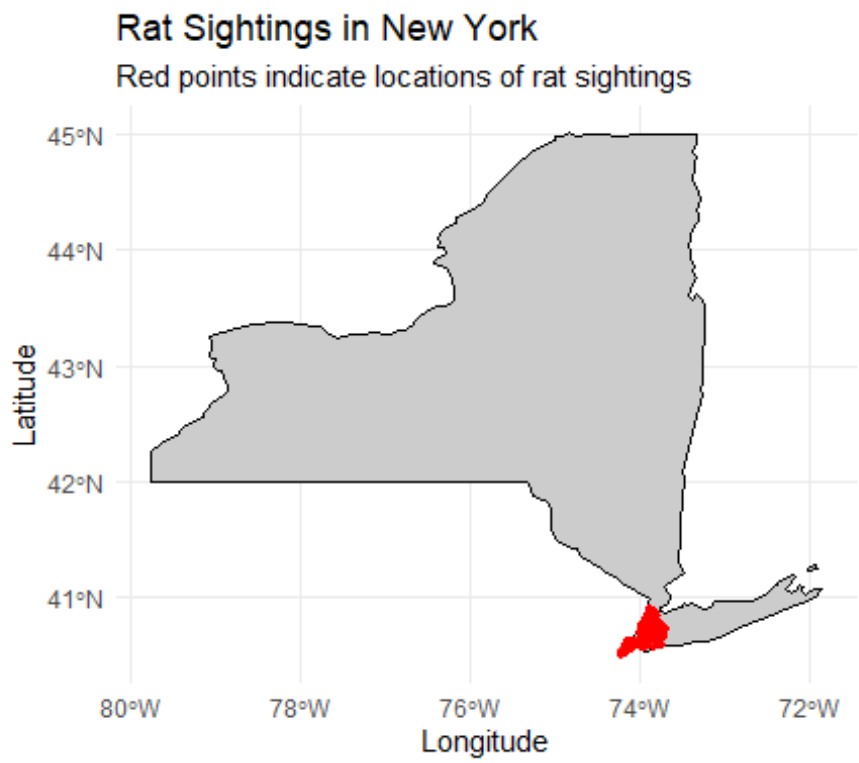
Contrast: The grey bars guarantee unobstructed clarity. Repetition: Easy comparisons are made possible by consistent formatting throughout years. Alignment: Labels and bars are arranged to make them easy to read. Proximity: Each year's data is arranged rationally

Implementation of the Principles of Heaven and Cairo Comparability, Year-to-year trend analysis is made possible by the faceted design. Clarity & Simplicity: Distractions are eliminated by a simple arrangement. Functionality: The graphic identifies important regions for focused pest management initiatives. 🚀

```
us_states <- read_sf("cb_2018_us_state_20m/cb_2018_us_state_20m.shp")

us_states_hires <-
read_sf("ne_10m_admin_1_states_provinces/ne_10m_admin_1_states_provinces.shp"
)
```

#Plot and Filter for New York

```
# Load US states shapefile
us_states <- st_read("cb_2018_us_state_20m/cb_2018_us_state_20m.shp")  # Make
sure the path is correct

## Reading layer `cb_2018_us_state_20m' from data source
##
`C:\Users\Password\Downloads\cb_2018_us_state_20m\cb_2018_us_state_20m.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 52 features and 9 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -179.1743 ymin: 17.91377 xmax: 179.7739 ymax:
71.35256
## Geodetic CRS:  NAD83

# Filter for New York
ny_state <- us_states %>% filter(NAME == "New York")

# Load rat sightings data (assuming it has latitude and longitude columns)
rats_data <- read.csv("Cleaned_A1_sightings.csv")  # Ensure this file is in
your working directory

# Convert rat sightings into an sf object
rats_sf <- st_as_sf(rats_data, coords = c("longitude", "latitude"), crs =
st_crs(ny_state))

# Plot the map with rat sightings
ggplot() +
  geom_sf(data = ny_state, fill = "grey80", color = "black") +  # New York
state map
  geom_sf(data = rats_sf, color = "red", alpha = 0.5, size = 0.7) +  # Rat
sightings
  labs(title = "Rat Sightings in New York",
       subtitle = "Red points indicate locations of rat sightings",
       x = "Longitude", y = "Latitude") +
  theme_minimal()
```




```
# Save as PNG
ggsave(filename = "output/A1(4).png", width = 10, height = 7, dpi = 300)

# Save as PDF
ggsave(filename = "output/A1 (4).pdf", width = 10, height = 7)

# Load US states shapefile
us_states <- st_read("cb_2018_us_state_20m/cb_2018_us_state_20m.shp")

## Reading layer `cb_2018_us_state_20m' from data source
##
`C:\Users\Password\Downloads\cb_2018_us_state_20m\cb_2018_us_state_20m.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 52 features and 9 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -179.1743 ymin: 17.91377 xmax: 179.7739 ymax:
71.35256
## Geodetic CRS:  NAD83

# Filter only for New York state
ny_state <- us_states %>% filter(NAME == "New York")

# Load rat sightings dataset
rats_data <- read_csv("A1_sightings.csv", na = c("", "NA", "N/A"),
show_col_types = FALSE)
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame
for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```r
# Check for missing values in latitude and longitude
rats_data %>%
  summarise(
    missing_longitude = sum(is.na(Longitude)),
    missing_latitude = sum(is.na(Latitude))
  )
```

```
## # A tibble: 1 × 2
##   missing_longitude missing_latitude
##               <int>            <int>
## 1               706              706
```

```r
# Remove rows with missing latitude/longitude
rats_data <- rats_data %>%
  drop_na(Longitude, Latitude) %>%
  mutate(
    longitude = as.numeric(Longitude),
    latitude = as.numeric(Latitude)
  )

# Convert rat sightings into an `sf` object
rats_sf <- st_as_sf(rats_data, coords = c("longitude", "latitude"), crs =
4326, remove = FALSE)

# Extract latitude and longitude explicitly
rats_sf <- rats_sf %>%
  mutate(
    latitude = st_coordinates(.)[, 2],  # Extract Y (latitude)
    longitude = st_coordinates(.)[, 1]  # Extract X (longitude)
  )

# Filter for NYC area using extracted coordinates
nyc_rats <- rats_sf %>%
  filter(latitude > 40.4 & latitude < 41.0, longitude > -74.3 & longitude < -
73.6)

# Plot refined NYC map with rat sightings
nyc_rat_map <- ggplot() +
  geom_sf(data = ny_state, fill = "grey80", color = "black", alpha = 0.5) +
# New York state outline
  geom_sf(data = nyc_rats, color = "red", alpha = 0.6, size = 1.5) +  # Rat
sightings
  labs(
    title = "NYC Rat Sightings - Improved Visualization",
```

```r
    subtitle = "Red points indicate rat complaint locations in New York
City",
    x = "Longitude", y = "Latitude"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16, color = "darkred"),
    plot.subtitle = element_text(size = 12, face = "italic"),
    axis.text = element_text(size = 10),
    legend.position = "none"
  ) +
  coord_sf(xlim = c(-74.3, -73.6), ylim = c(40.4, 41.0))  # Zoom into NYC
area

# Load US states shapefile
us_states <- st_read("cb_2018_us_state_20m/cb_2018_us_state_20m.shp")  #
Ensure correct file path
```

```
## Reading layer `cb_2018_us_state_20m' from data source
##
`C:\Users\Password\Downloads\cb_2018_us_state_20m\cb_2018_us_state_20m.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 52 features and 9 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -179.1743 ymin: 17.91377 xmax: 179.7739 ymax:
71.35256
## Geodetic CRS:  NAD83
```

```r
# Filter only for New York state
ny_state <- us_states %>% filter(NAME == "New York")

# Load rat sightings dataset
rats_data <- read_csv("A1_sightings.csv", na = c("", "NA", "N/A"),
show_col_types = FALSE)
```

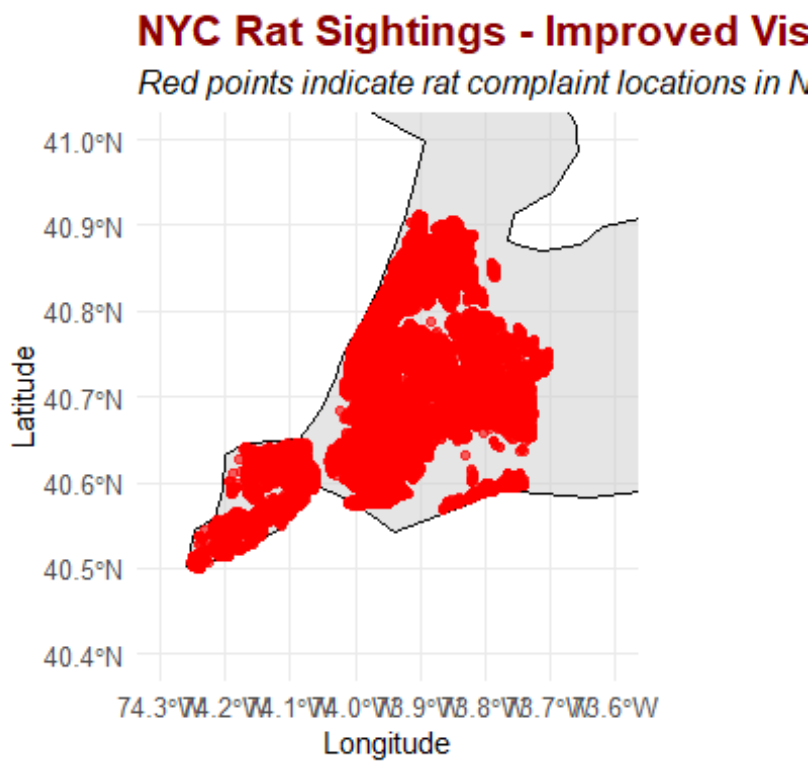
```
## Warning: One or more parsing issues, call `problems()` on your data frame
for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```r
# Check for missing values in latitude and longitude
rats_data %>%
  summarise(
    missing_longitude = sum(is.na(Longitude)),
    missing_latitude = sum(is.na(Latitude))
  )
```

```
## # A tibble: 1 × 2
##   missing_longitude missing_latitude
```

```
##                <int>             <int>
## 1                706               706

# Remove rows with missing latitude/longitude
rats_data <- rats_data %>%
  drop_na(Longitude, Latitude) %>%
  mutate(
    longitude = as.numeric(Longitude),
    latitude = as.numeric(Latitude)
  )

# Convert rat sightings into an `sf` object
rats_sf <- st_as_sf(rats_data, coords = c("longitude", "latitude"), crs =
4326, remove = FALSE)

# Extract latitude and longitude explicitly
rats_sf <- rats_sf %>%
  mutate(
    latitude = st_coordinates(.)[, 2],  # Extract Y (latitude)
    longitude = st_coordinates(.)[, 1]  # Extract X (longitude)
  )

# Filter for NYC area using extracted coordinates
nyc_rats <- rats_sf %>%
  filter(latitude > 40.4 & latitude < 41.0, longitude > -74.3 & longitude < -
73.6)

# Plot refined NYC map with rat sightings
nyc_rat_map <- ggplot() +
  geom_sf(data = ny_state, fill = "grey80", color = "black", alpha = 0.5) +
# New York state outline
  geom_sf(data = nyc_rats, color = "red", alpha = 0.6, size = 1.5) +  # Rat
sightings
  labs(
    title = "NYC Rat Sightings - Improved Visualization",
    subtitle = "Red points indicate rat complaint locations in New York
City",
    x = "Longitude", y = "Latitude"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16, color = "darkred"),
    plot.subtitle = element_text(size = 12, face = "italic"),
    axis.text = element_text(size = 10),
    legend.position = "none"
  ) +
  coord_sf(xlim = c(-74.3, -73.6), ylim = c(40.4, 41.0))  # Zoom into NYC
area
```

```r
# Display final map
print(nyc_rat_map)
```




```r
# Save as PNG
ggsave(filename = "output/A1(5).png", width = 10, height = 7, dpi = 300)

# Save as PDF
ggsave(filename = "output/A1(5).pdf", width = 10, height = 7)
```

REFLECTION

A geographic examination of rodent infestations throughout New York City is provided by the NYC Rat Sightings Map. The visualization aids in locating high-density infestation locations by superimposing a geographic border map with rat sighting data. By switching from a state-wide to a city-focused map, analysis is directed towards the places that require the greatest intervention.

CRAP • Contrast: Hot-spots are easily visible due to the grey map background's contrast with the red rat sightings.

• Repetition: Regular labeling and map design enhance readability and make comparing various zoom settings simple.

• Alignment: Rat sightings are accurately represented by aligning with NYC limits thanks to the logical structure of the geospatial data.

• Proximity: Only New York City is covered, avoiding distraction from other regions and guaranteeing that the data points are directly tied to their respective geographic locations.

There were two datasets used: Shapefiles from the United States are used to plot the boundaries of New York. Latitude and longitude coordinates are included in the NYC rat sightings data.

First, New York State was extracted by loading and filtering the shapefile with the U.S. state borders. The information of rat sightings was then transformed into a spatial (sf) object so that it could be plotted on a map.

To make sure the visualisation concentrates on metropolitan regions where rat sightings are most common, a bounding box filter was used to zoom in on NYC because the dataset encompasses the entirety of New York State.

Using the Visualisation Principles of Healy and Cairo Comparability (Healy): By directly comparing rat activity across boroughs, the map reveals patterns that a straightforward bar chart might miss. Clarity & Simplicity (Cairo): A minimalist approach with limited colors ensures that viewers immediately grasp the key insights without visual clutter. Truthfulness (Cairo): The visualization accurately represents data without distortion, ensuring the findings are credible and actionable. Functionality (Cairo & Healy): The map provides practical insights for urban pest management, helping authorities identify problem areas efficiently.